Prompting LLMs for Biological Research

Quick Reference · Write it, check it, verify it, log it

MDIBL CGDS Core
CGDS@mdibl.org

PROMPT SKELETON · LABELED OR PROSE, SAME INFORMATION

```
__TASK__
-
__CONTEXT__
- Organism:
- Sample / tissue:
- Assay / data type:
- Genome build / software versions:
__CONSTRAINTS__
- Do:
- Do not:
__OUTPUT_FORMAT__
-
__FALLBACK__
- If ..., return ... rather than guessing.
<data> ... </data>
```

```
I'd like you to [task, and what form the answer
should take]. This is from [organism], [tissue or
sample type], measured by [assay], using [genome
build / software versions]. Please [rules it should
follow], and don't [things it should avoid]. If
[something doesn't fit], tell me that rather than
guessing.

[pasted data]
```

Labels are a checklist for you, not a syntax the model needs. Paste the source material instead of relying on the model's memory, and explicitly allow it to say "unknown."

RUBRIC — CHECK EVERY PROMPT

COMPONENT	ASK YOURSELF
TASK	Did I say what I want done, starting with a verb?
CONTEXT	Every fact the model couldn't guess: organism, tissue, assay, versions?
FRAME	Did I say how it should reason?
AUDIENCE	Did I say who will read it?
STYLE / TONE	Did I say how it should read?
CONSTRAINTS	Rules it would otherwise break? Measurable, ~8 or fewer.
OUTPUT FORMAT	Exact shape, if it goes into code or a spreadsheet?
EXAMPLES	Would one worked example save a paragraph?
FALLBACK	What to do when input doesn't fit? "Unknown" over a guess?

BIOLOGY PITFALLS TO AVOID

Unstated organism	You'll get human biology.
Paralog collapse	Say symbols are mapped orthologs; flag merged paralogs.
Symbols only	Use stable IDs (Ensembl) and name the convention.
No build / versions	State genome build and package versions.
Memory recall	Never ask for citations or numbers from memory.
Story as analysis	Run the enrichment test first, then interpret.
Leading question	"If my interpretation isn't supported, say so first."

MATCH THE CHECK TO THE CLAIM

OUTPUT	CHECK AGAINST
Gene identity, function	Ensembl, NCBI Gene, UniProt, ZFIN / FlyBase / WormBase / MGI
Citations	Open via PubMed or DOI; confirm it says what's claimed
Enrichment claims	A formal enrichment test on your data
Code	Run on a small known dataset; check package docs
Numbers	Your data, protocol, datasheet, primer database
Extractions	Spot-check a random sample against the source

LOG EVERY RESEARCH PROMPT

Model + version	e.g. name, version, date accessed
Interface	Web chat / API / coding agent / local
Full prompt	Verbatim, version-controlled
Inputs	Exact file, version, or md5
Iterations	What changed and why, not just the final version
Validation	How the output was checked

Same prompt, different answer: LLMs sample each token, so outputs vary between runs. Treat every output as a draft to verify.